SMACK(T)
Sena, Mumtaz, Arushi, Karl, Williams Xu
Info 200 King
Section [BG] Gores
Group Envision Project
05/26/2023

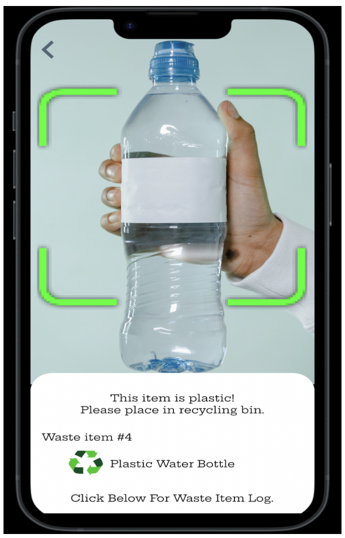
Draft Envision Project - Changeability: Waste Reducing App

We have become accustomed to a society where ease and comfort have been the main focal point for everything in life. Even if that comes at the expense of our planet. Climate Change is at the forefront of many debates. Often leaving the average human confused and unsure on what's causing this problem or how they can tackle it by small changes in their daily lives. Carbon emissions, overconsumption, and waste managemnt are the main issues that have been adding to climate change (NASA, 2022). Many people often have no idea what carbon emissions are; let alone the effect it has on our planet. When we are consuming, we are unaware of the waste we are actually producing and the catastrophic effect it will have on our planet. According to the Environmental Protection Agency, one of the main problems at hand is as individuals we have no way of knowing just how much waste consumption and carbon footprint we put out into the world and there isn't a way to independently track our carbon footprint, overconsumption, and waste management as daily contributors to this issue. The real issue at hand is how can we as individuals decrease our carbon footprint, waste management and overconsumption. The reason this is a problem for the average person is because they do not know what would fall under the category of carbon emissions, overconsumption and how one can manage their waste. They can simply google what it is everytime, but they have no real way of tracking their own carbon footprint/emissions which is the key issue here.

The average human in the United States of America has 16 tons of carbon footprint. Carbon footprint can simply equate the usage of electricity that comes from burning coal and using gasoline when we drive. Transportation by itself is 28 percent of carbon emissions. In 2005 in the United States alone, we had the highest carbon emission consumption which equated to 6.14 Billion tons of carbon. One ton equates to 2000 pounds. According to a study done by the University of Michigan, just food alone makes up 10-30 percent of one household's carbon footprint. According to the same study done by the University of Michigan; meats in one's diet alone makes up 56.6 percent of greenhouse gas emissions. There aren't tools that record one's diet to explain how much carbon footprint a product or activity has. Knowing that meat makes up more than half of one's greenhouse gas emission alone is one of the biggest wake up calls on the real issue at hand and that is how can an individual track their own carbon footprint and have the right tools to tackle this global issue at hand.

FUNCTIONALITY

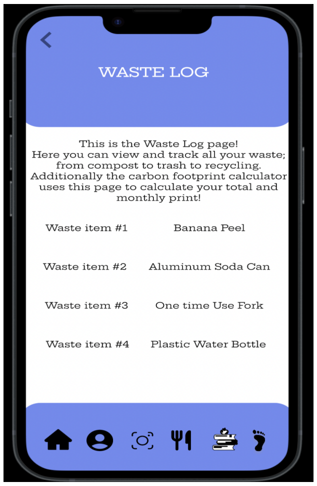
Changeability was developed to help reduce waste and change the user's habits. Climate change is a global issue. Over the last century, there was a drastic increase in carbon emissions and waste products due to the industrial revolution. The purpose of this app is to decrease the user's carbon footprint/emissions and waste while improving climate change and making the user more sustainable. The features that focus on resolving these issues are the waste scanner, waste log, food/diet, and carbon emissions. We will discuss each page's functions and how they solve our waste problem.



*This is the waste scanner page.

Waste scanner

The waste scanner page allows the user to scan their waste and log it. When the user scans the waste product, the algorithm will let the user know what type of waste it is and where it should be discarded. The problem that we are tackling is immense waste. One of the main contributors to climate change is waste. Many people are not aware of the damages caused by waste, nor do they recognize their accumulated waste. This leads to a lack of knowledge and consciousness when it comes to global and personal waste. This page is efficient and convenient as it makes the grueling and time-consuming task of manually entering your waste much quicker.



*This is the waste log page.

Waste log

The waste log page allows users to track their waste. In addition, this feature is significant to the carbon emissions page as it uses this and many more features to calculate the user's carbon footprint. This feature helps solve the problem by making the user conscious of their waste. This is known as the "Priming phenomenon," when we become aware of certain thoughts or behaviors that can lead to changes. This is significant because when the user enables these features, they begin to recognize their amount of waste. Once the user becomes conscious of their waste, they will start to make changes to their habits. As the user notices their amount of waste, the app will assist them in changing their habits and creating a more sustainable lifestyle.

*Food/diet (log) page.

Food/diet (log)

The food/diet page assists the user in creating healthier and sustainable food choices. This feature recommends sustainable foods and tells the user the carbon dioxide emission per kilogram. The food/diet page is essential to the app as it contributes to the calculation of the user's carbon footprint. This page has a similar feature to the waste log page as it tracks the user's food intake and total carbon emitted from the food.

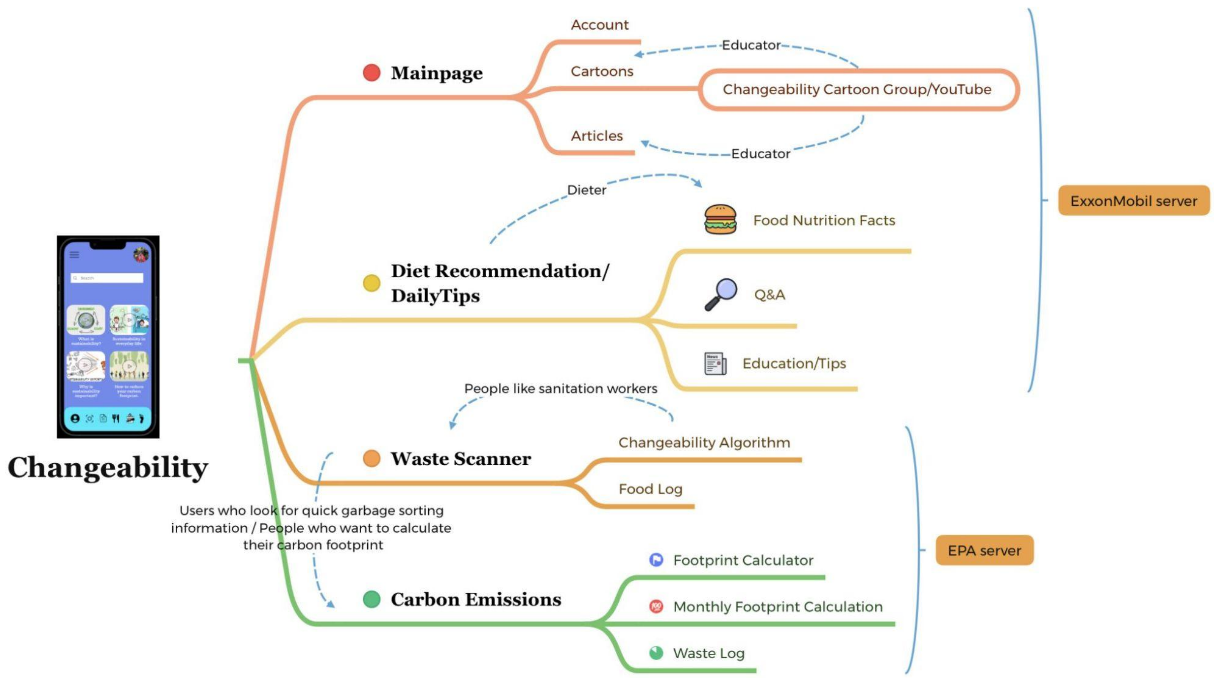
*These are the carbon emission pages.

Carbon emissions (Overall and Monthly)

The last features are the overall and monthly carbon footprint/emission pages. These pages help the users acknowledge and recognize their waste and unsustainable lifestyle by understanding their contribution to climate change. This page calculates and determines the user's carbon footprint score by collecting data such as the onboarding questions, waste log, and food/diet log. The overall carbon footprint feature assists the user in envisioning their overall contribution to climate change, while the monthly carbon footprint page gives the user a glimpse into what their waste pattern is. Furthermore, this option allows users to track and see their improvements over the months.

The features within changeability assist the users in understanding and acknowledging their daily waste consumption providing them with resources that will help them reduce their carbon footprint. Ultimately our goal is to facilitate a process where users can make changes to their habits while reducing global and personal waste. Changeability is an all-in-one sustainability app that seeks to improve the user's habits and the environment by enabling convenient and beneficial features such as a waste scanner, waste log, food/diet log, and carbon footprint tracker. The ability to change for sustainability is within us all, and by utilizing the functions in this app, the users and the environment will improve and benefit from it.

ARCHITECTURE



This diagram depicts the architecture of the Changeability information system. All the key features include in the application are introduced as follow:

- **Changeability app**, is a sustainability assistance app that is available on both IOS and Android. It can assist users to identify the correct category of the waste. It can also calculate the carbon footprint users produced and give tips depending on their questions and statistics. There are educational cartoons and healthy diet instructions provided to help users learn more about what carbon emissions are, what's the effect to the environment by the waste people produce, and what they can do personally to improve world sustainability. This app also has questions for each user in order to distinguish and send the relevant information due to their diet habits.
- **Users**, who search for quick garbage sorting information, calculate their own carbon footprint statistics and study global health knowledge. Users can get tips and track their monthly carbon footprint. Teachers can also be one of the potential users to teach students about the category of different waste with the help of the quick response.
- **Changeability server**, after the login, there are questions available to help to allocate different resources and transit information and videos from EPA, ExxonMobil and YouTube. Statistics will be collected for monthly calculation use.
- **EPA server**, stands for United States Environmental Protection Agency, serves as a database that provides all the data and analysis we need currently for this app, including the different amount of carbon footprint of each product. It also provides carbon footprint calculation. Moreover, there is more than carbon footprint in the database, for example,

the average temperature for each state in America every year, which is available for the future update of Changeability.

- **ExxonMobil server**, provides relevant sustainability articles on the mainpage, responses for Q&A, and reference for diet recommendation and education tips users search for.
- **Changeability app & YouTube**, Changeability itself has a group of people who will work on creating videos, including cartoons, that are accessible to all different age groups about sustainability topics, like waste recycling, and global environment situation, etc. Changeability will also cooperate with and select videos from Youtube with related topics.
- **Changeability algorithm**, Changeability will develop its own algorithm on the function of the waste scanner. It’s an artificial intelligence system that will automatically process the photo by going through light adjustment, exposure, key part cutting and focus, etc. With the help of the database, EPA, it can eventually provide the users its decision.

POLICY

When looking at pertinent policies for our app, one that was most relevant to our focus was **RCW 70A.45.020**, from the Washington State Legislature. This policy describes the requirements behind Washington’s greenhouse gas emissions reductions, listing the years in which overall emissions must be lowered to a specified point, one requirement being that emissions must be ninety-five percent below 1990 levels by 2050. Our envisioned app design contributes to this policy greatly, with the aforementioned carbon emissions pages allowing individuals to properly track their emissions and the educational content of the application providing users with the necessary information to accomplish their sustainability goals, which in turn affects the statewide greenhouse gas emissions.

Other important policies are those outlined within the **Apple App Store Review Guidelines** and the **Google Play Developer Policy Center**. These lay out the proper requirements and guidelines to be able to publish applications onto the respective app stores. While our application would of course attempt to fulfill every section of the guidelines, one area in particular that can be addressed is that related to data privacy. **Section 5.1.1 (iii)** of Apple’s App Store Review Guidelines outlines the aspect of Data Minimization, which is that apps should only ever “…request access to data relevant to the core functionality of the app…” (Apple, 2022). Changeability takes this into consideration as the only user data it requires is one’s email address, a password, and the self-input food/waste log. The app does not utilize personal data beyond that point, thus falling in line with the guidelines.

BENEFITS AND HARMS

Changeability creates a net positive impact for the consumer:

Benefits:

- Context is established by generating output based on a **variety of factors** like diet, transportation, and family size, which are also factors utilized in the Footprint Calculator Organization's website (Global Footprint Network, 2023). These elements are analyzed holistically for a more nuanced analysis of an individual's habits.
-There is an element of **personalization** by allowing for the creation of a user profile, letting an individual track their change over time and get a sense of their developing habits. Making a profile controls how information is presented to and accessed by the user (Lozhkin, 2011). It also allows the user to compare their new patterns to their old patterns in a concrete, quantitative way. Changeability's waste log functionality provides a clear record of everything the user has disposed of, and provides data for the in-app carbon footprint calculator.
-There are **multimodal educational opportunities** for constructive improvement. Some benefits of multimedia learning include more effectively presenting abstract concepts and stimulating a greater interest in learning (Abdulrahaman, M, et al., 2020). The app includes simple, introductory videos on sustainability and carbon footprint. There is also a page for overall analysis that gives gentle prompting feedback when the user performs below under expectations. This takes away the shame associated with making mistakes, and provides several methods to take in information, increasing accessibility.
-The app **simplifies decision making** by allowing the user to scan their waste product and be told where it goes. Takes away the information overload of recycling/composting and waste policies that ultimately lead a lot of people to make mistakes when getting rid of their waste.
-Changeability **addresses several facets of a sustainability journey** - education, performance, feedback, and monitoring, all in a loop. One of the app pages recommends healthy, sustainable food, which oftentimes is also more environmentally friendly in its emissions and packaging, leading to a cycle of healthy living that has a minimal impact on our surroundings.

Despite Changeability addressing a majority of the focus problems, there are some unintentional negative effects that it creates:

Harms:

- Changeability uses a variety of factors in calculating carbon footprint, but still **fails to capture significant sources of emissions** like plane flights, and home energy efficiency. This means that there may be a disconnect between a person's perception of their carbon footprint and their true carbon footprint.

-The app has a **limited scope** - carbon footprint is just one of the many factors that we contribute to harming the environment, and this app makes it seem like by decreasing our carbon footprint we are resolving all sustainability issues. Other concerns that could be addressed, possibly in the education section of the app, include activities like overfishing, deforestation, pollution, and using harmful chemicals. A more accurate scope would be to focus on ecological footprint, which gives a better picture of human impact on the ecological resources we utilize (Ecological Footprint, 2020).

- The app **de-emphasizes the importance of non-technological ways to combat waste production**, like legal or societal activism, and community-level behavior changes. Several high level initiatives like the Renewable Energy Program and New Source Performance Standards for emissions are behind the progress made on sustainability improvements (EPA, 2023).
- Changeability contributes to existing **data privacy issues** by failing to address a privacy policy ensuring sensitive information remains secure (Tobin, 2021). An app that tracks waste may also as a side effect be able to track what a consumer is buying, thereby revealing their age demographic, shopping preferences, and other personal information that may be sold to third parties unless said otherwise.

References

1. “How Many Planets Does It Take to Sustain Your Lifestyle?” Ecological Footprint Calculator, 2023. https://www.footprintcalculator.org/home/en.
2. Lozhkin, Igor. “What Is User Profile? Why Is It Important? How to Implement It?” Arnica, June 24, 2011. https://www.arnicasoftware.com/blog/what-is-user-profile-why-is-it-important-how-to-implement-it-/2825/index.aspx.
3. Abdulrahaman, M D, N Faruk, A A Oloyede, et al. “Multimedia Tools in the Teaching and Learning Processes: A Systematic Review.” Heliyon, November 2, 2020. https://www.sciencedirect.com/science/article/pii/S2405844020321551.
4. “Ecological Footprint.” WWF, 2020. https://wwf.panda.org/discover/knowledge_hub/all_publications/ecological_footprint2/.
5. “Climate Change Regulatory Actions and Initiatives.” EPA, 2023. https://www.epa.gov/climate-change/climate-change-regulatory-actions-and-initiatives.
6. Tobin, Donal. “What Is Data Privacy-and Why Is It Important?” Integrate.io, May 7, 2021. https://www.integrate.io/blog/what-is-data-privacy-why-is-it-important/.
7. NASA. (2023, June 1). *Carbon dioxide concentration*. NASA. https://climate.nasa.gov/vital-signs/carbon-dioxide/#:~:text=Key%20Takeaway%3A,in%20less%20than%20200%20years.
8. Environmental Protection Agency. (2022, December 3). *The Current National Picture*. EPA. https://www.epa.gov/facts-and-figures-about-materials-waste-and-recycling/national-overview-facts-and-figures-materials
9. Inc., Apple. (2022). *App Store Review Guidelines*. Apple Developer. https://developer.apple.com/app-store/review/guidelines/#data-collection-and-storage
10. Google. (2023). *Developer policy center*. Google. https://play.google.com/about/developer-content-policy/
11. Washington State Legislature. *Greenhouse gas emissions reductions—Reporting requirements.* RCW 70A.45.020: Greenhouse gas emissions reductions-reporting requirements. (2021). https://apps.leg.wa.gov/rcw/default.aspx?cite=70A.45.020
12. SMACK(T). (2023). SMACK(T) Info 200 Group Project - Changeability [figma design].https://www.figma.com/file/shGs4G2Tbpk3to4I6ssnB0/INFO-200-GROUP-PROJECT-SMACK(T)--sustainability-app?type=design&node-id=0%3A1&t=2jJZT62Z5wIufYpL-1
13. *Carbon Footprint Factsheet*. Center for Sustainable Systems. (n.d.). https://css.umich.edu/publications/factsheets/sustainability-indicators/carbon-footprint-factsheet
14. Ritchie, H., Roser, M., & Rosado, P. (2020, May 11). *CO2 emissions*. Our World in Data. https://ourworldindata.org/co2-emissions